

Tu 13

$$\lambda^2 - 6\lambda + 13 = 0$$

$$(\lambda - 3)^2 + 4 = 0$$

$$\lambda - 3 = \pm 2i$$

$$\lambda = 3 \pm 2i$$

Name

- For the ODE $y'' - 6y' + 13y = e^{-2t}$

- Write down the complementary sol.

$$y_c(t) = c_1 e^{3t} \cos(2t) + c_2 e^{3t} \sin(2t)$$

- Compute a particular sol.

$$y_p(t) = \frac{1}{29} e^{-2t}$$

- The Aug Mat for $y(0) = 2, y'(0) = 3$ is

$$y_p = a e^{-2t}$$

$$y_p' = -2a e^{-2t}, y_p'' = 4a e^{-2t}$$

$$(4a + 12a + 13a) e^{-2t} = e^{-2t}$$

$$29a = 1$$

$$y = y_c + y_p$$

$$y(0) = \frac{1}{29} e^0 + c_1 e^0 \cos(0) + c_2 e^0 \sin(0) = 2$$

$$\frac{1}{29} + c_1 + 0 c_2 = 2$$

$$c_1 = 2 - \frac{1}{29}$$

1. For the ODE $y'' - 5y' - 14y = \sin(t)$

- 1.1. Write down the complementary sol.

$$y_c(t) =$$

- 1.2. Compute a particular sol.

$$y_p(t) =$$

- 1.3. The Aug Mat for $y(0) = 1, y'(0) = 4$ is

- For the ODE $y'' - 4y' + 9y = 3\sin(2t)$ with

- Write down the complementary sol.

$$y_c(t) =$$

- Compute a particular sol.

$$y_p(t) =$$

- The Aug Mat for $y(0) = 2, y'(0) = -3$ is

2. For the ODE $y'' - 2y' + 5y = 2\cos(3t)$

- 2.1. Write down the complementary sol.

$$y_c(t) =$$

- 2.2. Compute a particular sol.

$$y_p(t) =$$

- 2.3. The Aug Mat for $y(0) = -2, y'(0) = 7$ is